ModelIt!

Student Activity Pack

The Wildfire Feedback Loop

Modeling How Fire, Climate, and Ecosystems Create a Self-Reinforcing Cycle

Grade 09L1 | NGSS HS-ESS3-3, HS-LS2-6 | Stability and Change

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Why do you think wildfires in the western United States have been getting larger and more destructive in recent decades?

2 Is fire always bad for a forest? Can you think of reasons fire might actually be necessary for a healthy ecosystem?

3 What factors do you think determine how fast a wildfire spreads and how hot it burns?

4 Draw a diagram showing what you think happens to a forest ecosystem after a major wildfire — include what changes over the next 5, 20, and 50 years.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Wind Speed | Fuel Load | Fire Intensity | Soil Moisture | Vegetation Recovery

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Healthy Fire Cycle

Low wind, moderate fuel, adequate soil moisture

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Catastrophic Fire

High wind, maximum fuel, drought conditions

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Feedback Loop Test

Track system behavior over multiple fire cycles

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Carbon Release — The massive amount of CO_2 released when vegetation burns — a single large wildfire can release more carbon in weeks than the surrounding forest absorbed in decades, contributing to climate change that makes future fires worse

Fire Weather Index — A composite measure of temperature, humidity, wind speed, and drought that predicts fire danger — climate change is increasing the number of high fire-weather days per year in fire-prone regions

Pyrocumulonimbus Events — Extreme wildfires that generate their own thunderstorms, producing lightning that starts new fires and erratic winds that make the original fire unpredictable — a feedback loop within the feedback loop, representing the most dangerous fire behavior on Earth

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why are wildfires getting bigger, hotter, and harder to stop — and is the cycle feeding itself? Use evidence from your simulation to support your answers.

1. **How does the positive feedback loop in your model explain why wildfires are getting progressively worse over time?**

2. **What does your model reveal about why a century of fire suppression has been counterproductive?**

3. **How does climate change interact with the wildfire feedback loop to accelerate the cycle?**

4. **If low-intensity fire is beneficial, how should forest management balance fire suppression with ecological fire needs?**

5. **What are the limitations of modeling wildfire behavior with only five components when real fires involve hundreds of variables?**

8 STEM Challenge: Design a Community Wildfire Resilience Plan

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YOUR ENGINEERING MISSION

Design a comprehensive wildfire management plan for a community in the wildland-urban interface that breaks the feedback loop through prevention, preparation, and ecosystem restoration.

THE SCENARIO

A mountain community of 15,000 people borders a national forest that hasn't experienced a major fire in 80 years. Fuel loads are at record levels, drought conditions are worsening, and the forest is surrounded by new housing developments. The community has hired your team to design a wildfire resilience plan that addresses both immediate safety and long-term ecosystem health. Your plan must balance fire prevention with the ecological need for fire.

Engineering Design Thinking

1. WHAT WOULD YOUR DEFENSIBLE SPACE REQUIREMENTS LOOK LIKE FOR HOMES WITHIN 1 MILE OF THE FOREST BOUNDARY?

2. HOW WOULD YOU DESIGN A PRESCRIBED BURN PROGRAM THAT REDUCES FUEL LOADS WITHOUT EXCEEDING SAFE CONDITIONS?

3. WHAT EARLY WARNING AND EVACUATION SYSTEMS WOULD YOUR PLAN INCLUDE, AND HOW WOULD THEY ACCOUNT FOR WIND SPEED AND DIRECTION CHANGES?

4. HOW WOULD YOU RESTORE THE NATURAL FIRE CYCLE IN THE SURROUNDING FOREST WHILE PROTECTING THE COMMUNITY?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

Real Career Connection

Wildfire Scientists study fire behavior, ecology, and management strategies using computational models and field research, earning $55,000–$110,000/year. Fire Ecologists study how fire shapes ecosystems and design prescribed burn programs, earning $50,000–$100,000/year. Urban Planners specializing in wildfire resilience design communities that can coexist with fire-prone landscapes, earning $60,000–$120,000/year.